Audio Signal Recognition Based on Intervals' Numbers (INs) Classification Techniques

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***Abstract*—Speech is one of the major human-machine interaction modalities and it is especially important in the case of special education using social robots. Although modern speech recognition engines can effectively deal with normal human-robot conversations, there are instances in special education where additional word detection and word comparison capabilities are needed to run in parallel with the typical conversation flow. This paper investigates the efficiency of a word detection method based on intervals' numbers.**

Keywords—Speech recognition, Human-machine interaction, Intervals' numbers (IN)

I. Introduction

Scientists in the field of socially intelligent robotics focus on developing robots capable of exhibiting natural-appearing social qualities [1]. Beyond the basic capabilities of moving and acting autonomously, social robots need to communicate with humans in a social and engaging manner. Dialogue is the main joint process of communication [2], [3]. Humans employ a variety of linguistic social cues (i.e. words or phrases that reveal sympathy, empathy, encouragement etc.) to regulate a dialogue. These cues would also be effective for controlling human-robot interaction, which is important due to the current robots' perception (e.g. speech processing) limits [4]. Thus, in order to improve the quality of human-robot interaction, it is essential for social robots to be capable of guiding a dialogue towards specific directions, so as, for example, to serve specific therapeutic intervention purposes.

Towards this end, the methodology presented in this paper is intended to serve the latter purpose; contribute towards the design of therapeutic interventions for children with learning disabilities or Autism Spectrum Disorder (ASD) where speech is the basic communication channel of child-robot interaction. More specifically, in this work trigger words to regulate a conversation and guide the therapeutic activities are represented by Interval Numbers (INs). An IN is an established mathematical object [5], [6] that may represent either a fuzzy interval or a distribution of numbers. A proposed k nearest neighbor (kNN) classification scheme is adopted to recognize the trigger words. This method is especially designed to be a computationally inexpensive tool used in conjunction with the robots' build-in speech recognition engine.

The proposed technique falls into Lattice Computing (LC) paradigm. LC techniques are expected to be beneficial to human-robot interaction applications, since they can deal with both numerical and non-numerical data (e.g. verbal cues) and facilitate computations with semantics. Moreover, LC techniques can deal with fuzzy sets, engage logic and reasoning and can process data fast [7], [8].

Speech recordings may result in big data, therefore efficient data representation and processing are essential. The engagement of INs in this work provides significant data reduction without losing data statistics. No prepossessing or feature extraction takes place; the INs scheme results directly from the raw data (i.e. the speech signal).

The layout of the paper is as follows: Section II briefly outlines the theoretical background of LC and INs. Section III describes the proposed experimental methods. Section IV presents the experimental results. Section V concludes by discussing the contribution of the paper and the related upcoming work.

II. Theoretical Background

In this section, the hierarchy of lattices from Level-0 to Level-2, resulting to the lattice of INs [5], [6], [9] is presented.

A. Application of INs

The application of INs to Neural Networks (NNs) and Fuzzy Inference Systems (FIS) have been reported in the literature [6], [8], [10]–[12].

In [6], an LC extension of a fuzzy neural classifies for human facial expression recognition was presented. The reported advantages of working with INs are the incremental real-time learning and generalization based on fuzzy lattice reasoning (FLR) techniques. The proposed classifier performs efficiently, demonstrating the capability of INs to represent partially ordered non-numeric and nonstationary data (i.e. data semantics). The effectiveness of INs in demonstrated in [10]; accommodation of granular data, introduction of tunable non-linearities and induction of descriptive decision-making knowledge from the data, are some of the additionally reported advantages of INs application in three different benchmark problems. In [11] INs represent a distribution of image features for the recognition of image patterns. An FLR classifier is developed to learn distributions of INs, and four well-known benchmark image pattern recognition applications are demonstrated. In [12] novel schemes for tunable fuzzy rule interpolation and extrapolation are presented, based on INs. Preliminary application involves the reconstruction of partially occluded human facial expressions based on a NN.

In all aforementioned applications of INs, the algorithm CALCIN is used for inducing an IN from a distribution. Despite its proven performance, CALCIN has been met with criticism because of the fact that the induced IN does not retain a clear statistical interpretation. An extension of CALCIN, namely algorithm distrIN, was proposed in response, in order to provide a straightforward, statistical interpretation [8]. The pilot application of distrIN that was presented in [8], uses electroencephalography (EEG) signals with significant data reduction without losing the statistical features of the data. The kNN classification scheme was used to demonstrate the effectiveness of the newly proposed technique, resulting in

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good classification performances compared to alternative state-of-the-art methods in the literature.

B. Lattice Hierarchy

This section presents a hierarchy of lattices, resulting in the lattice of INs. The analytic theory of FLR is presented in [5], [6], [11].

A lattice is a partially ordered set $(L, \sqsubseteq)$ with the property that any two elements $x, y \in L$ have both a greatest lower bound (infimum) represented by $x \sqcap y$ and a least upper bound (supremum) represented by $x \sqcup y$. If in a lattice $(L, \sqsubseteq)$ every pair (x, y) satisfies either $x \sqsubseteq y$ or $x \sqsupset y$ then the lattice $(L, \sqsubseteq)$ is totally-ordered. A lattice $(L, \sqsubseteq)$ is called complete if each of its subsets X has both an infimum and a supremum in L. A non-void complete lattice contains both a least element and a greatest element symbolized by o and i, respectively [5].

1) Level-0; the lattice $(\bar{R}, \leq)$ of Real Numbers: Consider the totally-ordered complete lattice $(\bar{R}, \leq)$, where $\bar{R} = R \cup \{-\infty, +\infty\}$ with least and greatest elements $-\infty$ and $+\infty$, respectively. The corresponding infimum and supremum operations are symbolized by $\wedge$ and $\vee$, respectively.

2) Level-1; the lattice $(I, \subseteq)$ of Type-1 intervals: Considering $a_1, a_2 \in \bar{R}$ such that $a_1 \leqslant a_2$, a Type-1 interval $A = [a_1, a_2]$ is defined as $[a_1, a_2] = \{x : x \in \bar{R} \text{ and } a_1 \leqslant x \leqslant a_2\}$. The set of Type-1 intervals in $\bar{R}$ (including the empty interval) is symbolized by I_1. The lattice $(I_1, \subseteq)$ is a complete lattice, where $\subseteq$ is the set theoretic inclusion. Considering two intervals $A = [a_1, a_2]$ and $B = [b_1, b_2]$ in I_1, their infimum $(\cap)$ is given by $A \cap B = a_1 \vee b_1, a_2 \wedge b_2$, if and only if $a_1 \vee b_1 \leq a_2 \wedge b_2$, and their supremum $(\dot{\cup})$ is given by $A \dot{\cup} B = a_1 \wedge b_1, a_2 \vee b_2$.

Given of (1) a positive valuation function v: $L \rightarrow R_0^+$, and (2) a dual isomorphic function θ: $L \rightarrow L$ in lattice $(L, \leq)$, results a positive valuation function v_1: $L \times L \rightarrow R_0^+$ in lattice $(L \times L, \geq \times \leq)$ of *generalized intervals*, is given by $v_1([a,b]) = v(\theta(a)) + v(b)$. Therefore, both a metric function $d_1(.,.)$ and two fuzzy order functions $\sigma_\sqcap(.,.)$ and $\sigma_\sqcup(.,.)$, can be defined. The following three functions are available in $(I_1, \sqsubseteq)$.

The metric function d_1: $I_1 \times I_1 \rightarrow R_0^+$ in lattice $(I_1, \sqsubseteq)$ is calculated as follows:

$$d_1([a,b],[c,e]) = [v(\theta(a \wedge c)) - v(\theta(a \vee c))] + [v(b \vee e) - v(b \wedge e)] \quad (1)$$

The two fuzzy order functions are introduced as $\sigma_\sqcap(x, y) = \frac{v(x \sqcap y)}{v(x)}$, namely σ-meet, and $\sigma_\sqcup(x, y) = \frac{v(y)}{v(x \sqcup y)}$, namely σ-join.

Parametric functions $v(.)$ and $\theta(.)$ can be selected with various ways and may introduce tunable nonlinearities towards optimal classification schemes.

3) Level-2; The lattice $(F_1, \preccurlyeq)$ of INs: A Generalized Intervals' Number (GIN) is defined as a function f: $[0,1] \rightarrow (\bar{R} \times \bar{R}, \geq \times \leq)$ where $(\bar{R} \times \bar{R}, \geq \times \leq)$ is a lattice of generalized intervals. Assume G the set of GINs. Then, the $(G, \sqsubseteq)$ is a complete lattice. The interest is focused to the sublattice of IN. An IN, is defined as a function F: $[0,1] \rightarrow I_1$, which satisfies the following:

- $h_1 \leq h_2 \Rightarrow F_{h_1} \sqsupseteq F_{h_2}$ and
- $\forall X \subseteq [0,1]$: $\bigcap_{h \in X} F_h = F_{\vee X}$

The set F_1 of INs is a partially ordered, complete lattice, which is denoted as $(F_1, \preccurlyeq)$. An IN is interpreted as a grain of information [5]. It has been shown that the set F_1 is a metric lattice with cardinality equal to the infinity $\aleph_1$ of set R of the real numbers. In other words, there are so many INs, as real numbers.

The proposed distrIN algorithm is presented in Fig.1. Assume a speech signal recording as shown in Fig.1(a). All the different speech signal values of Fig.1(a) can form a set of $n+1$ sorted real numbers $x_0 < x_1 < \ldots < x_n$. i.e. a distribution function. An IN, equivalently, can be represented as a set of intervals F_h, $h \in [0,1]$, which is the interval-representation, shown in Fig.1(b), or as a function $F(x) = \bigvee_{h \in [0,1]} \{h : x \in F_h\}$, which is the membership-function-representation, shown in Fig.1(c). As expected, audio signals representing the same spoken word produce very similar distributions, even for different speaker voices. However, different spoken words produce audio signals which result in more varied distribution shapes, and are therefore distinguishable. Note that the membership-function-representation of the induced IN is identical to the corresponding distribution function. This is a straightforward representation with a clear statistical interpretation. An IN is represented by L intervals all having the same end, where only one of the intervals is trivial by construction. In conclusion, it takes L real numbers to represent an IN. An IN also represents a distribution, i.e. orders of magnitude more than L.

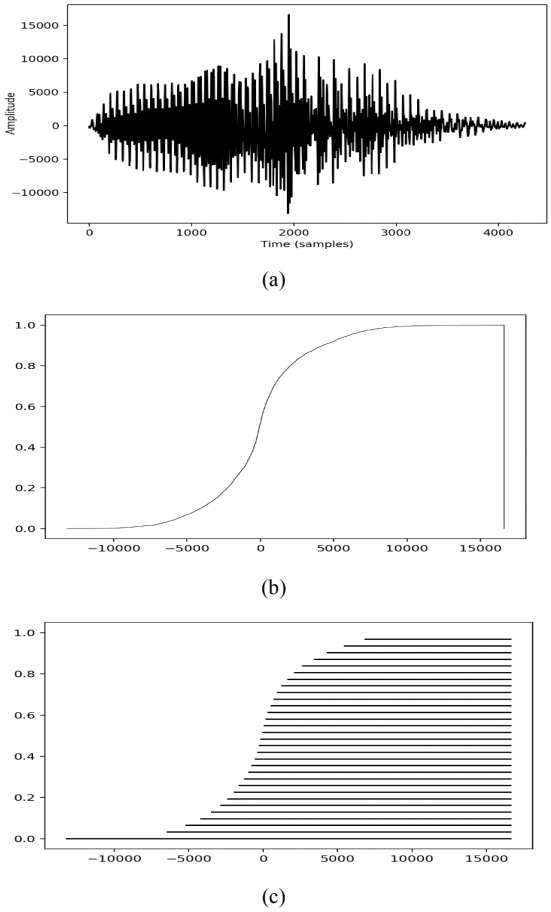


Fig. 1. Induction of IN F from a speech signal. (a) the speech signal corresponds to the recording of a spoken digit. (b) The membership-function-representation of the induced IN F. (c) The interval-representation of IN F for L=32 cuts

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III. EXPERIMENTAL METHODS

The word classification experiments were conducted using the spoken digits open dataset found in [13]. The dataset contains 2000 recordings of English utterances of digits 'zero' to 'nine' by four speakers. The audio signals were extracted from the audio files and were converted to INs, each described by 32 cuts, as described in the previous section. This resulted in the creation of 2000 INs, i.e. 200 INs per class.

A. Classification

For the classification tasks, the kNN classification method was employed with a 'leave-one-out' approach. This approach was selected because of its reliability, given that the amount of our training data was not prohibitively large. Table I shows how the INs are arranged for a classification task.

TABLE I. ARRANGEMENT OF INS FOR THE CLASSIFICATION TASK

	List of INs		Label
Subject1	$IN_{1,1}$	$IN_{1,32}$	ZERO
⋮	⋮	⋮	⋮
Subject200	$IN_{200,1}$	$IN_{200,32}$	ZERO
⋮	⋮	⋮	⋮
Subject1801	$IN_{1801,1}$	$IN_{1801,32}$	NINE
⋮	⋮	⋮	⋮
Subject2000	$IN_{2000,1}$	$IN_{2000,32}$	NINE

Initially, a subject IN is selected as the test datum, and the rest of the subjects are marked as training data. The test subject and each of the training subjects are then compared using the distance metric or one of the two fuzzy order functions presented in the previous section. Having selected one of the three metrics, it is used to compare all of the corresponding cuts of the two INs. The mean of these per-cut comparisons, is a representation of the overall relationship between the two INs. The relationships between the test subject and the training subjects are then sorted in order to find the k INs most related to the test subject. The sorting order depends on the metric used; Training INs that are most related to the test subject are those with the shortest distance (closest to zero) but the highest fuzzy order value (closest to 1). The overall nearest neighbour is then determined using majority voting and the classes of the nearest neighbour and the test subject are compared to determine if classification was successful. The process is repeated until all INs have been classified.

B. Optimization

The classification rate depends on the parameters chosen for the parametric functions $v(x)$ and $\theta(x)$. For this purpose, a genetic algorithm has been implemented. In this GA, each individual is a set of $v(x)$ and $\theta(x)$ parameters, as well as the k value used in the kNN classifier.

An initial population of 100 individuals (sets of parameters) is generated. In each generation, every individual's set of parameters are used in a classification task as described above, in order to evaluate the parameters' performance. The performance of each individual is described by a vector D, which contains the diagonal elements of the resulting confusion matrix. The cost C of an individual is calculated using the mean of vector D, subtracted by the optimal performance of 100%, i.e. $C = 100\text{-}mean(D)$. To improve the speed of evaluation of the individuals, parallel processing techniques were employed.

Evaluation of all individuals results in a l ist of costs, used to rank the individuals' performance and genetic operations are applied. Crossover is applied in every generation by exchanging random sets of genes between randomly selected individuals. Mutation is applied every five generations by modifying the values of randomly selected parameters of a predefined number of individuals.

IV. EXPERIMENTAL RESULTS

The experiments that were carried out investigated the classification performance of the proposed method for the following sets of classes: a) three classes (spoken digits '0', '5' and '9'), b) five classes (spoken digits '0', '3', '5', '7' and '9') and c) ten classes for all spoken digits. In each case, the distance metric and the two fuzzy order functions were evaluated. In addition, the effect in performance of employing the linear and sigmoid functions, with and without parameters, as the $v(x)$ and $\theta(x)$ parametric functions for the calculation of the relationship between pairs of INs was also investigated.

Initially, the classification experiments were carried out with no parameters for the linear ($v(x) = x$ and $\theta(x) = -x$) and sigmoid ($v(x) = 1/(1 + e^{-x})$ and $\theta(x) = -1/(1 + e^{-x})$) functions.

For the kNN classifier, an arbitrary value of 4 for k was selected. The classification results for the distance metric and the two fuzzy order functions using the linear and sigmoid functions with default parameters are presented in Table II.

TABLE II. CLASSIFICATION RESULTS WITH NO PARAMETERS

Metric	Parametric function	3 classes	5 classes	10 classes
Distance	Linear	70.5%	51.3%	34.45%
	Sigmoid	54.5%	38.8%	24.05%
Join	Linear	64.5%	52.7%	39.8%
	Sigmoid	41.83%	23.4%	16.85%
Meet	Linear	67.33%	55%	41.5%
	Sigmoid	41.83%	23.4$	16.1%

Following the classification results with the default parameters, the parameter optimization process took place. Parameters were then introduced to the parametric equations in order to create tunable distance and fuzzy order functions, which would alter the comparisons between the INs, and would in turn lead to different classification accuracies. The positions of the parameters in the parametric equations are shown in (3) and (4) for the linear type and in in (5) and (6) for the sigmoid.

$$v(x) = v_0 x \tag{3}$$

$$\theta(x) = -\theta_0 x \tag{4}$$

$$v(x) = v_0/(1 + \exp(-v_1 x - v_2)) + v_3 \tag{5}$$

$$\theta(x) = -\theta_0/(1 + \exp(-\theta_1 x - \theta_2)) + \theta_3 \tag{6}$$

For each of the six conditions (the three sets of classes and the two types of parametric functions), a GA was executed. The conditions for the GA was an initial population of 100 individuals and the duration of the algorithm was 10 generations. The classification rate for the best individual and the corresponding parameters were recorded. The classification results are shown in Table III.

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TABLE III. CLASSIFICATION RESULTS AFTER OPTIMISATION

Metric	Parametric function	3 classes	5 classes	10 classes
Distance	Linear	81%	70.7%	54%
	Sigmoid	59.66%	40.2%	25.7%
Join	Linear	70.16%	58.4%	43.1%
	Sigmoid	60.83%	36.9%	16.85%
Meet	Linear	70%	56.9%	41.7%
	Sigmoid	53.66%	34%	18.85%

It can be seen from the experimental results that the use of linear parametric functions is significantly better than the sigmoid for the particular classification problem. In this case the nature of the audio signals, is such that this type of parametric function is more suitable for INs' comparison.

It is also evident that the GA improves the classification performance considerably. It appears that appropriate parameter selection is vital to the effectiveness of the proposed method. These results are a clear indication that more work is needed on the selection of appropriate parametric functions according to the characteristics of the original signal.

V. DISCUSSION

The experimental results presented in this paper demonstrate the potential of the proposed methodology. The observed classification rates illustrate that spoken words can be distinguished from each other, even though there is no pre-processing of the audio signals and no feature extraction techniques are employed, which implies a reduced computational burden and a less complex algorithm. Also, the introduction of tunable non-linearities through the parametric functions $v(.)$ and $\theta(.)$ and the subsequent use of a GA for their optimization, leads to significantly improved performance. Another major advantage of the method is the compression achieved without the loss of data statistics. The main disadvantage of INs is the loss of the time dimension. Future work will consider the preservation of the time dimension by computing INs per time-windows for a time series, as well as and the use of other classification schemes.

The methodology presented in this paper is intended to contribute towards the design of therapeutic interventions for children with certain types of learning disabilities and children with ASD as part of the national project titled "Social Robots as Tools in Special Education", where speech will be a vital component of child-robot interaction. This method is aimed to be a computationally inexpensive tool used in conjunction with the robots' built-in speech recognition engine to recognize trigger words, as well as a detection mechanism for speech sound errors observed in children with learning disabilities. The generalization properties of the method allow the investigation of such varied tasks that will be pursued for the purposes of the project.

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